

EDUCATION

Monash University
Bachelor's Double Degree — Mechatronics Engineering / Commerce (majoring in Finance)

- Coursework includes microcontroller programming, circuit analysis, and foundational commerce units

Melbourne, Australia
Expected 2029

PROJECTS

Multi-Node Wireless Sensor Network
Personal embedded systems project — ESP32, Flask, SQLite

- Designed and built a three-node environmental sensing network on ESP32 dev boards using BME280 (temperature/humidity/pressure), PIR motion, and LDR light sensors
- Implemented an HTTP POST data pipeline from each node into a Flask/SQLite backend, with stale-node detection and a live web dashboard
- Investigating an MQTT upgrade as a stretch goal, with a planned written comparison against the HTTP-based approach

2026

Ultrasonic Distance Meter
Personal project — Arduino Uno, C

- Built an Arduino-based distance measurement system using an HC-SR04 ultrasonic sensor with real-time output on a 16x2 LCD display
- Measured distances to four decimal places of precision, with readings triggered by a pushbutton

2026

SKILLS

Programming: Python, C

Embedded / Hardware: STM32, ESP32, Arduino Uno, sensor integration (BME280, PIR, LDR, HC-SR04), circuit analysis

Web / Backend: Flask, SQLite, HTML/CSS, REST APIs, MQTT (in progress)

Tools: Git/GitHub, Make, Linux/command line

Languages: English (native), German (conversational, ongoing study)